```
2303
```

```
import sys
!{sys.executable} -m pip install transformers datasets torch pandas scikit-learn
print("Libraries installed successfully.")
```

```
Requirement already satisfied: scikit-learn in /usr/local/lib/python3.12/dist-packages (1.6.1)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from transformers) (3.25.0)
Requirement already satisfied: huggingface-hub<2.0,>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (1.5
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2.0.2)
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (26.0)
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from transformers) (6.0.3)
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12/dist-packages (from transformers) (2025.11.3)
Requirement already satisfied: tokenizers<=0.23.0,>=0.22.0 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.2
Requirement already satisfied: typer-slim in /usr/local/lib/python3.12/dist-packages (from transformers) (0.24.0)
Requirement already satisfied: safetensors>=0.4.3 in /usr/local/lib/python3.12/dist-packages (from transformers) (0.7.0)
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-packages (from transformers) (4.67.3)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (18.1.0)
Requirement already satisfied: dill<0.3.9,>=0.3.0 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.3.8)
Requirement already satisfied: requests>=2.32.2 in /usr/local/lib/python3.12/dist-packages (from datasets) (2.32.4)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from datasets) (3.6.0)
Requirement already satisfied: multiprocess<0.70.17 in /usr/local/lib/python3.12/dist-packages (from datasets) (0.70.16)
Requirement already satisfied: fsspec<=2025.3.0,>=2023.1.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025
Requirement already satisfied: typing-extensions>=4.10.0 in /usr/local/lib/python3.12/dist-packages (from torch) (4.15.0)
Requirement already satisfied: setuptools in /usr/local/lib/python3.12/dist-packages (from torch) (75.2.0)
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dist-packages (from torch) (1.14.0)
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/dist-packages (from torch) (3.6.1)
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packages (from torch) (3.1.6)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.3)
Requirement already satisfied: scipy>=1.6.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.16.3)
Requirement already satisfied: joblib>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (1.5.3)
Requirement already satisfied: threadpoolctl>=3.1.0 in /usr/local/lib/python3.12/dist-packages (from scikit-learn) (3.6.0)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]<=2025.
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->
Requirement already satisfied: typer in /usr/local/lib/python3.12/dist-packages (from huggingface-hub<2.0,>=1.3.0->transformer
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.17
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->dat
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets) (3.11
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.32.2->datasets)
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.12/dist-packages (from sympy>=1.13.3->torch) (1.3.
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/dist-packages (from jinja2->torch) (3.0.3)
Requirement already satisfied: aiohappyeyeballs>=2.5.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0
Requirement already satisfied: aiosignal>=1.4.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fssp
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->f
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: anyio in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub<2.0,>=
Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hu
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->hug
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.
Requirement already satisfied: shellingham>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,
Requirement already satisfied: rich>=12.3.0 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.0,>=1.3.
Requirement already satisfied: annotated-doc>=0.0.2 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub<2.
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->hug
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich>=12.3.0->typer->h
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich>=12.3.0
Libraries installed successfully.
```

```
from datasets import load_dataset

# Load the 'imdb' dataset
imdb_dataset = load_dataset('imdb')

# Print the dataset to inspect its structure
print(imdb_dataset)
print("Dataset loaded successfully.")
```

```
/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: UserWarning:
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings tab (https://huggingface.co/settings/tokens), set it
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access public models or datasets.
  warnings.warn(
Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to enable higher rate limits and faster d
WARNING:huggingface_hub.utils._http:Warning: You are sending unauthenticated requests to the HF Hub. Please set a HF_TOKEN to en
```

README.md: 7.81k/? [00:00<00:00, 466kB/s]

plain_text/train-00000-of-00001.parquet: 100% 21.0M/21.0M [00:00<00:00, 105MB/s]

plain_text/test-00000-of-00001.parquet: 100% 20.5M/20.5M [00:00<00:00, 102MB/s]

plain_text/unsupervised-00000-of-00001.p(…): 100% 42.0M/42.0M [00:00<00:00, 109MB/s]

Generating train split: 100% 25000/25000 [00:00<00:00, 102736.44 examples/s]

Generating test split: 100% 25000/25000 [00:00<00:00, 86400.38 examples/s]

Generating unsupervised split: 100% 50000/50000 [00:00<00:00, 120226.70 examples/s]

```
DatasetDict({
    train: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    test: Dataset({
        features: ['text', 'label'],
        num_rows: 25000
    })
    unsupervised: Dataset({
        features: ['text', 'label'],
        num_rows: 50000
    })
})
Dataset loaded successfully.
```

```
from transformers import DistilBertTokenizerFast

# 1. Load the DistilBertTokenizerFast tokenizer
tokenizer = DistilBertTokenizerFast.from_pretrained('distilbert-base-uncased')

# 2. Define a tokenization function
def tokenize_function(examples):
    return tokenizer(examples['text'], truncation=True, padding=True)

# 3. Apply this tokenization function to all splits of the imdb_dataset
tokenized_datasets = imdb_dataset.map(tokenize_function, batched=True, remove_columns=['text'])

print("Dataset tokenization complete.")
print(tokenized_datasets)
```

tokenizer_config.json: 100% 48.0/48.0 [00:00<00:00, 2.12kB/s]

vocab.txt: 232k/? [00:00<00:00, 5.01MB/s]

tokenizer.json: 466k/? [00:00<00:00, 9.29MB/s]

Map: 100% 25000/25000 [00:32<00:00, 818.01 examples/s]

Map: 100% 25000/25000 [00:27<00:00, 879.99 examples/s]

Map: 100% 50000/50000 [01:02<00:00, 830.14 examples/s]

```
Dataset tokenization complete.
DatasetDict({
    train: Dataset({
        features: ['label', 'input_ids', 'attention_mask'],
        num_rows: 25000
    })
    test: Dataset({
        features: ['label', 'input_ids', 'attention_mask'],
        num_rows: 25000
    })
    unsupervised: Dataset({
        features: ['label', 'input_ids', 'attention_mask'],
        num_rows: 50000
    })
})
```

```
import torch

# 1. Extract the 'train' split
train_dataset = tokenized_datasets['train']

# 2. Extract the 'test' split (which is usually referred to as evaluation set in HuggingFace)
eval_dataset = tokenized_datasets['test']

# 3. Print the structure of both datasets
print("Train Dataset Structure:", train_dataset)
print("\nEvaluation Dataset Structure:", eval_dataset)
print("Datasets prepared successfully.")
```

```
Train Dataset Structure: Dataset({
    features: ['label', 'input_ids', 'attention_mask'],
    num_rows: 25000
})

Evaluation Dataset Structure: Dataset({
    features: ['label', 'input_ids', 'attention_mask'],
    num_rows: 25000
})
Datasets prepared successfully.
```

```
from transformers import DataCollatorWithPadding
from torch.utils.data import DataLoader

# 1. Set the format of both train_dataset and eval_dataset to 'torch'
train_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask', 'label'])
eval_dataset.set_format(type='torch', columns=['input_ids', 'attention_mask', 'label'])

# 2. Instantiate DataCollatorWithPadding
data_collator = DataCollatorWithPadding(tokenizer=tokenizer)

# 3. Create DataLoader instances
batch_size = 8 # Adjust based on GPU memory
train_dataloader = DataLoader(train_dataset, shuffle=True, batch_size=batch_size, collate_fn=data_collator)
eval_dataloader = DataLoader(eval_dataset, batch_size=batch_size, collate_fn=data_collator)

print("PyTorch Datasets and DataLoaders created successfully.")
print(f"Train DataLoader batch size: {train_dataloader.batch_size}")
print(f"Evaluation DataLoader batch size: {eval_dataloader.batch_size}")
```

```
PyTorch Datasets and DataLoaders created successfully.
Train DataLoader batch size: 8
Evaluation DataLoader batch size: 8
```

```
from transformers import DistilBertForSequenceClassification

# 1. Load the pre-trained DistilBertForSequenceClassification model
# 2. Specify num_labels=2 for binary sentiment classification (positive/negative)
model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased', num_labels=2)

print("Pre-trained DistilBertForSequenceClassification model loaded successfully.")
# 3. Optionally, print the model architecture to verify
print(model)
```

config.json: 100% 483/483 [00:00<00:00, 38.0kB/s]

model.safetensors: 100% 268M/268M [00:01<00:00, 231MB/s]

Loading weights: 100% 100/100 [00:00<00:00, 412.07it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]

```
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.weight | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.weight       | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Pre-trained DistilBertForSequenceClassification model loaded successfully.
DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
          (attention): DistilBertSelfAttention(
            (q_lin): Linear(in_features=768, out_features=768, bias=True)
            (k_lin): Linear(in_features=768, out_features=768, bias=True)
            (v_lin): Linear(in_features=768, out_features=768, bias=True)
            (out_lin): Linear(in_features=768, out_features=768, bias=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
            (lin1): Linear(in_features=768, out_features=3072, bias=True)
            (lin2): Linear(in_features=3072, out_features=768, bias=True)
            (activation): GELUActivation()
          )
          (output_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
        )
      )
    )
  )
  (pre_classifier): Linear(in_features=768, out_features=768, bias=True)
  (classifier): Linear(in_features=768, out_features=2, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
)
```

```python
from transformers import DistilBertForSequenceClassification

# 1. Load the pre-trained DistilBertForSequenceClassification model
# 2. Specify num_labels=2 for binary sentiment classification (positive/negative)
model = DistilBertForSequenceClassification.from_pretrained('distilbert-base-uncased', num_labels=2)

print("Pre-trained DistilBertForSequenceClassification model loaded successfully.")
# 3. Optionally, print the model architecture to verify
print(model)
```

```
Loading weights: 100%                    100/100 [00:00<00:00, 426.49it/s, Materializing param=distilbert.transformer.layer.5.sa_layer_norm.weight]
DistilBertForSequenceClassification LOAD REPORT from: distilbert-base-uncased
Key                     | Status     |
------------------------+------------+-
vocab_layer_norm.weight | UNEXPECTED |
vocab_projector.bias    | UNEXPECTED |
vocab_layer_norm.bias   | UNEXPECTED |
vocab_transform.bias    | UNEXPECTED |
vocab_transform.weight  | UNEXPECTED |
pre_classifier.weight   | MISSING    |
pre_classifier.bias     | MISSING    |
classifier.weight       | MISSING    |
classifier.bias         | MISSING    |

Notes:
- UNEXPECTED    :can be ignored when loading from different task/architecture; not ok if you expect identical arch.
- MISSING   :those params were newly initialized because missing from the checkpoint. Consider training on your downstream task.
Pre-trained DistilBertForSequenceClassification model loaded successfully.
DistilBertForSequenceClassification(
  (distilbert): DistilBertModel(
    (embeddings): Embeddings(
      (word_embeddings): Embedding(30522, 768, padding_idx=0)
      (position_embeddings): Embedding(512, 768)
      (LayerNorm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
      (dropout): Dropout(p=0.1, inplace=False)
    )
    (transformer): Transformer(
      (layer): ModuleList(
        (0-5): 6 x TransformerBlock(
          (attention): DistilBertSelfAttention(
            (q_lin): Linear(in_features=768, out_features=768, bias=True)
            (k_lin): Linear(in_features=768, out_features=768, bias=True)
            (v_lin): Linear(in_features=768, out_features=768, bias=True)
            (out_lin): Linear(in_features=768, out_features=768, bias=True)
            (dropout): Dropout(p=0.1, inplace=False)
          )
          (sa_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
          (ffn): FFN(
            (dropout): Dropout(p=0.1, inplace=False)
            (lin1): Linear(in_features=768, out_features=3072, bias=True)
            (lin2): Linear(in_features=3072, out_features=768, bias=True)
            (activation): GELUActivation()
          )
          (output_layer_norm): LayerNorm((768,), eps=1e-12, elementwise_affine=True)
        )
      )
    )
  )
  (pre_classifier): Linear(in_features=768, out_features=768, bias=True)
  (classifier): Linear(in_features=768, out_features=2, bias=True)
  (dropout): Dropout(p=0.2, inplace=False)
)
```

```
from transformers import TrainingArguments

# 2. Instantiate TrainingArguments
training_args = TrainingArguments(
    output_dir='./results',
    num_train_epochs=3,
    per_device_train_batch_size=8,
    per_device_eval_batch_size=8,
    warmup_steps=500,
    weight_decay=0.01,
    logging_dir='./logs',
    logging_steps=10,
    # Temporarily removing evaluation_strategy and save_strategy due to TypeError,
    # and disabling load_best_model_at_end to resolve the subsequent ValueError.
    # Ideally, these would be set to 'epoch' and match.
    load_best_model_at_end=False, # Changed to False to bypass ValueError
    metric_for_best_model='accuracy',
    report_to='none'
)

print("Training Arguments defined successfully.")
# 3. Print the instantiated TrainingArguments object to verify the settings
print(training_args)
```

```
`logging_dir` is deprecated and will be removed in v5.2. Please set `TENSORBOARD_LOGGING_DIR` instead.
Training Arguments defined successfully.
TrainingArguments(
```

```
accelerator_config={'split_batches': False, 'dispatch_batches': None, 'even_batches': True, 'use_seedable_sampler': True, 'non_
adam_beta1=0.9,
adam_beta2=0.999,
adam_epsilon=1e-08,
auto_find_batch_size=False,
average_tokens_across_devices=True,
batch_eval_metrics=False,
bf16=False,
bf16_full_eval=False,
data_seed=None,
dataloader_drop_last=False,
dataloader_num_workers=0,
dataloader_persistent_workers=False,
dataloader_pin_memory=True,
dataloader_prefetch_factor=None,
ddp_backend=None,
ddp_broadcast_buffers=None,
ddp_bucket_cap_mb=None,
ddp_find_unused_parameters=None,
ddp_timeout=1800,
debug=[],
deepspeed=None,
disable_tqdm=False,
do_eval=False,
do_predict=False,
do_train=False,
enable_jit_checkpoint=False,
eval_accumulation_steps=None,
eval_delay=0,
eval_do_concat_batches=True,
eval_on_start=False,
eval_steps=None,
eval_strategy=IntervalStrategy.NO,
eval_use_gather_object=False,
fp16=False,
fp16_full_eval=False,
fsdp=[],
fsdp_config={'min_num_params': 0, 'xla': False, 'xla_fsdp_v2': False, 'xla_fsdp_grad_ckpt': False},
full_determinism=False,
gradient_accumulation_steps=1,
gradient_checkpointing=False,
gradient_checkpointing_kwargs=None,
greater_is_better=True,
group_by_length=False,
hub_always_push=False,
hub_model_id=None,
hub_private_repo=None,
hub_revision=None,
hub_strategy=HubStrategy.EVERY_SAVE,
hub_token=<HUB_TOKEN>,
ignore_data_skip=False,
include_for_metrics=[],
include_num_input_tokens_seen=no,
label_names=None,
```

```
import sys
!{sys.executable} -m pip install evaluate
print(" 'evaluate' library installed successfully.")
```

```
Collecting evaluate
  Downloading evaluate-0.4.6-py3-none-any.whl.metadata (9.5 kB)
Requirement already satisfied: datasets>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (4.0.0)
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.0.2)
Requirement already satisfied: dill in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.3.8)
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.2.2)
Requirement already satisfied: requests>=2.19.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (2.32.4)
Requirement already satisfied: tqdm>=4.62.1 in /usr/local/lib/python3.12/dist-packages (from evaluate) (4.67.3)
Requirement already satisfied: xxhash in /usr/local/lib/python3.12/dist-packages (from evaluate) (3.6.0)
Requirement already satisfied: multiprocess in /usr/local/lib/python3.12/dist-packages (from evaluate) (0.70.16)
Requirement already satisfied: fsspec>=2021.05.0 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]>=2021.05.0->evalu
Requirement already satisfied: huggingface-hub>=0.7.0 in /usr/local/lib/python3.12/dist-packages (from evaluate) (1.5.0)
Requirement already satisfied: packaging in /usr/local/lib/python3.12/dist-packages (from evaluate) (26.0)
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (3.25.0)
Requirement already satisfied: pyarrow>=15.0.0 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (18.1
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-packages (from datasets>=2.0.0->evaluate) (6.0.3)
Requirement already satisfied: aiohttp!=4.0.0a0,!=4.0.0a1 in /usr/local/lib/python3.12/dist-packages (from fsspec[http]>=2021.05
Requirement already satisfied: hf-xet<2.0.0,>=1.2.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.7.0->eva
Requirement already satisfied: httpx<1,>=0.23.0 in /usr/local/lib/python3.12/dist-packages (from huggingface-hub>=0.7.0->evaluat
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Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluate) (
```

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Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.12/dist-packages (from requests>=2.19.0->evaluate) (
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Requirement already satisfied: attrs>=17.3.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fsspec
Requirement already satisfied: frozenlist>=1.1.1 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fs
Requirement already satisfied: multidict<7.0,>=4.5 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->
Requirement already satisfied: propcache>=0.2.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fss
Requirement already satisfied: yarl<2.0,>=1.17.0 in /usr/local/lib/python3.12/dist-packages (from aiohttp!=4.0.0a0,!=4.0.0a1->fs
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Requirement already satisfied: httpcore==1.* in /usr/local/lib/python3.12/dist-packages (from httpx<1,>=0.23.0->huggingface-hub>
Requirement already satisfied: h11>=0.16 in /usr/local/lib/python3.12/dist-packages (from httpcore==1.*->httpx<1,>=0.23.0->huggi
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas->evaluat
Requirement already satisfied: click>=8.2.1 in /usr/local/lib/python3.12/dist-packages (from typer->huggingface-hub>=0.7.0->eval
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Downloading evaluate-0.4.6-py3-none-any.whl (84 kB)
   ━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━━ 84.1/84.1 kB 5.1 MB/s eta 0:00:00
Installing collected packages: evaluate
Successfully installed evaluate-0.4.6
 'evaluate' library installed successfully.
```

```python
import evaluate
import numpy as np

# Define a function to compute evaluation metrics
def compute_metrics(eval_pred):
    predictions, labels = eval_pred
    # Apply np.argmax to predictions to get the predicted class labels.
    predictions = np.argmax(predictions, axis=1)

    # Load the metrics
    accuracy_metric = evaluate.load('accuracy')
    precision_metric = evaluate.load('precision')
    recall_metric = evaluate.load('recall')
    f1_metric = evaluate.load('f1')

    # Compute each metric
    accuracy = accuracy_metric.compute(predictions=predictions, references=labels)['accuracy']
    precision = precision_metric.compute(predictions=predictions, references=labels, average='binary')['precision']
    recall = recall_metric.compute(predictions=predictions, references=labels, average='binary')['recall']
    f1 = f1_metric.compute(predictions=predictions, references=labels, average='binary')['f1']

    return {
        'accuracy': accuracy,
        'precision': precision,
        'recall': recall,
        'f1': f1
    }

print("Compute metrics function defined successfully.")
```

```
Compute metrics function defined successfully.
```

```python
from transformers import Trainer

# Initialize the Trainer
trainer = Trainer(
    model=model,
    args=training_args,
    train_dataset=train_dataset,
    eval_dataset=eval_dataset,
    compute_metrics=compute_metrics,
    data_collator=data_collator,
)

print("Trainer initialized successfully.")
```

```
Trainer initialized successfully.
```

```
print("Starting model training...")
trainer.train()
print("Model training complete.")

print("Starting model evaluation...")
results = trainer.evaluate()
print("Model evaluation complete.")
```